

Electricity

ELECTRICITY IS THE PHENOMENON ASSOCIATED WITH EITHER MOVING OR STATIC ELECTRIC CHARGE.

SEE ALSO

◀ 113 Reactivity

◀ 148–149 Electrochemistry

◀ 168–169 Inside atoms

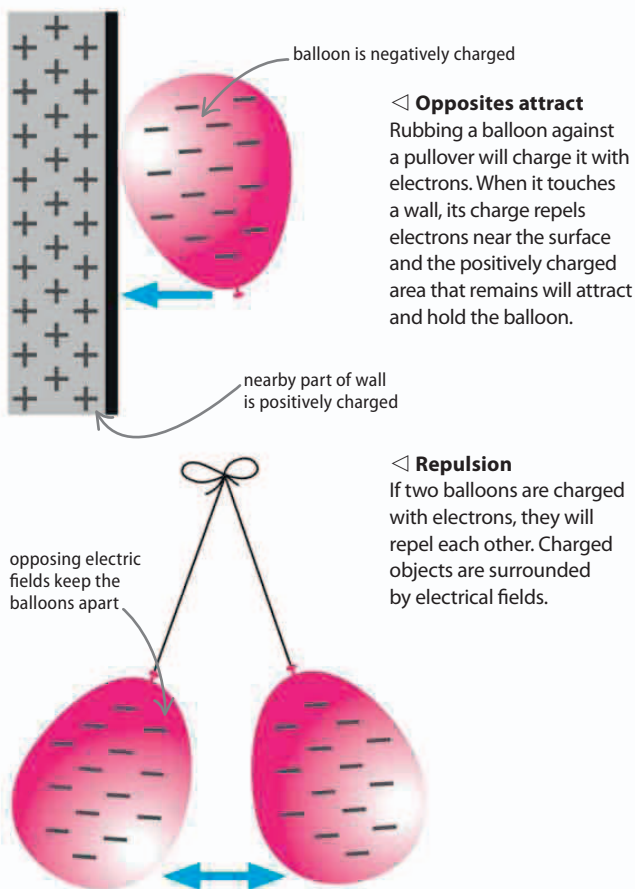
Circuits

206–207 ▶

Atoms contain tiny particles called electrons that carry negative electrical charge. These orbit the positively charged atomic nucleus, but can become detached.

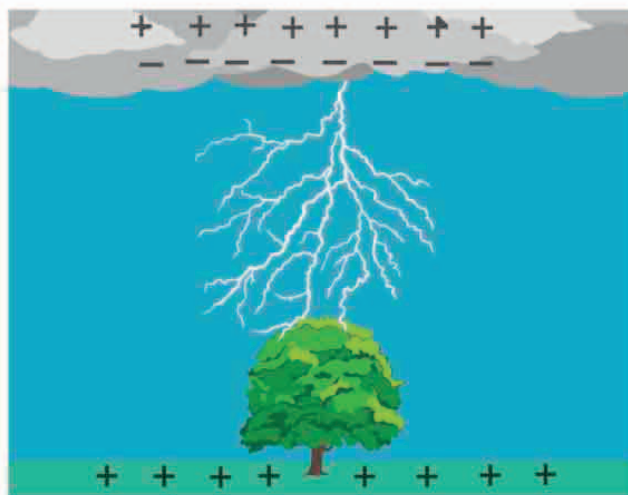
Static electricity

When an object contains an excess of electrons, it is said to be negatively charged. It will repel other negatively charged objects. Objects containing many atoms that have lost electrons are positively charged. Such objects attract negatively charged objects, and repel other positively charged objects. Since the electrons are not flowing to or from such objects, this type of electricity is called static. Objects with static charge also attract neutral objects, by repelling electrons within them to leave an area of positive charge.



Static discharge

In stormy weather, electrons gradually move from the Earth to low clouds. Charges also separate within clouds. The ground and the upper parts of clouds become strongly positively charged, while the lower parts of clouds become strongly negative. Eventually, the clouds discharge as the charges neutralize each other. Discharges within clouds are seen as sheet lightning, while forked lightning is a cloud-to-ground discharge. The lightning can travel at a staggering 209,200 km/h (130,000 mph) with an electric current of around 300,000 amperes.



△ Dangerous places

When lightning strikes, it takes the shortest route to the ground. A lone tree is likely to be struck in a thunderstorm. High buildings are often struck, too, so they are fitted with lightning rods to conduct any lightning safely down to the ground.

Lightning can heat the air surrounding it to a temperature that is **more than five times hotter than the Sun's surface!**